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Engr 095 FYRE Program Pathway 1: Sensing the World
Reflection 3

The past few weeks in this module have been mainly working on our final project of this half of course. We were given a presentation on how to present and create meaningful and enjoyable presentations and the importance of being able to give an effective presentation. Then we were given our project and given time to plan out our idea and somewhat of a plan to start our project before we began. We decided to use an umbrella to protect dogs. Then we were in the keysight lab and given all our time to create our prototypes come to life. We worked with bread boards, resistors, our PDOT sensors we made, and used AI to write code for our project to actually work. We also used building materials like cardboard, hot glue, duct tape, straws, pipe cleaners to make it a real prototype and look like a dog house. We first started off using online resources and AI to figure out how to configure the wires so all our elements were hooked up correctly to be able to use and get data from, like the PDOT sensor, servo motor, button, and photoresistor. After we figured that out we prompted ChatGPT to make code in micro python to send to the arduino microcontroller and be able to get the motor to move under the right conditions. Then we made the house out of cardboard and fit all the electronics inside and made it look nice. Our presentation was also something we worked on together to make it look nice and be effective at showing our project after we had a successful demo showing our prototype works correctly. Finally we presented and graded and reflected on others' presentations as well.

I learned a lot from this final section of the module. It taught me a lot on how it is working with a group on a more complicated hands-on project that is a really useful experience for the future. I also realized how much planning a big project before starting can help make me feel more organized and make it seem like a task that is more manageable by planning and talking out the process. It also introduced me to the world of circuits a lot more, I think it's really cool and I might be interested in taking a real course that would properly introduce me into how the breadboard and wires and electronics work more. Also throughout this whole project, AI was both a blessing and burden, sometimes giving us exactly what we wanted and more and other times it would get confused and give us differing answers so I learned more about where AI can help and where it can struggle and the importance of having good prompts to give it. The prompt can drastically affect what output you get and what it gives you and that it even differs between different AI models. Finally this was my first real presentation I have given in college and gave me a real experience of what it is and should be like now going forward and learned a lot of tips when making powerpoints that I will take with me into the future. This was a really impactful project and a really great experience and I am very grateful for it.